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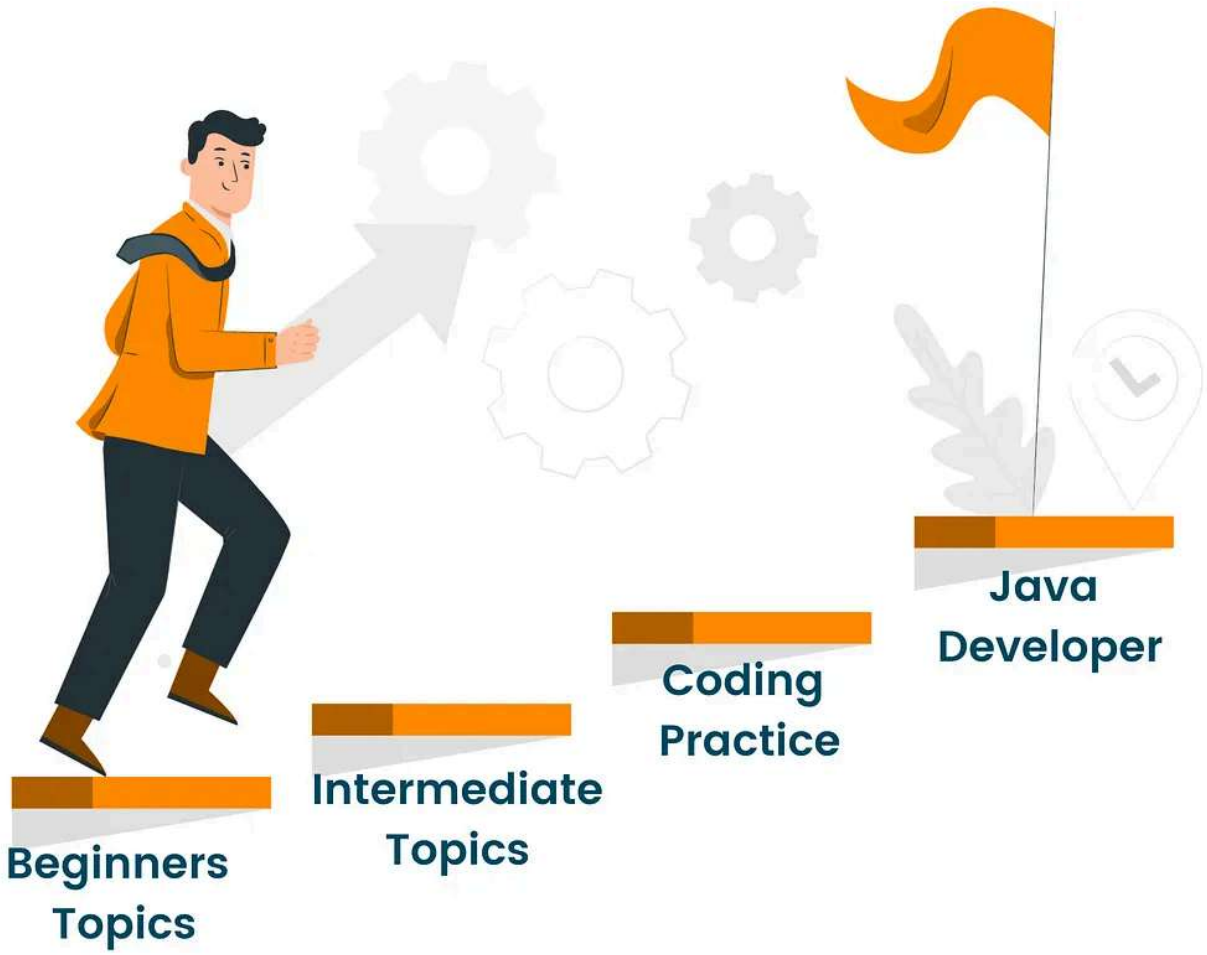

Complete Core Java Developer Roadmap with all Topics.

1. Introduction to Java

Amarjeet Kumar Singh Follow 4 min read · Aug 25, 2024

- **Overview of Java**: History, features, and use cases.
- **Setting Up Java Environment**: Install JDK, setup IDE (Eclipse, IntelliJ IDEA, or VS Code).
- **Hello World Program**: Writing your first Java program.

2. Basic Syntax and Structure

- **Java Syntax**: Keywords, identifiers, literals.
- **Identifiers**: Naming conventions, rules for identifiers.
- **Data Types**: Primitive types (int, char, boolean, etc.) and reference types.
- **Variables**: Declaration, initialization, and scope.
- **Operators**: Arithmetic, relational, logical, bitwise, assignment, and miscellaneous operators.
- **Control Flow Statements**:
- **Conditional Statements**: if, else if, else, switch-case.
- **Loops**: for, while, do-while, enhanced for-loop.
- **Keywords**: Understanding special keywords like `return`, `break`, `continue`, `instanceof`.

3. Arrays

- **Introduction to Arrays**: Single-dimensional and multi-dimensional arrays.
- **Array Operations**: Traversing, sorting, and searching arrays.

4. Strings and String Manipulation

- **String Class**: Creating and manipulating strings.

- **StringBuffer**: Mutable strings.
- **StringBuilder**: Fast mutable strings.

5. Methods

- **Defining Methods**: Method declaration, parameters, return values.
- **Method Overloading**: Same method name with different parameters.
- **Method Parameters**: Pass-by-value, varargs.

6. Type Casting

- **Primitive Type Casting**: Implicit and explicit casting.
- **Reference Type Casting**: Upcasting and downcasting.

7. Wrapper Classes

- **Wrapper Classes**: Understanding `Integer`, `Double`, `Character`, `Boolean`, etc.
- **Autoboxing and Unboxing**: Automatic conversion between primitive types and wrapper classes.

8. Core Classes

- **Object Class**: Methods provided by `Object` class like `toString()`, `equals()`, `hashCode()`.
- **Math Class**: Utility methods for mathematical operations.

9. Object-Oriented Programming (OOP)

- **Classes and Objects**: Defining classes, creating objects, constructors.
- **Constructors**: Default constructor, parameterized constructor, constructor overloading.

- **Inheritance**: extends keyword, method overriding, `super` keyword.
- **Polymorphism**: Compile-time (method overloading), runtime (method overriding), upcasting, and downcasting.
- **Abstraction**: Abstract classes and interfaces.
- **Encapsulation**: Access modifiers (private, protected, public, default).
- **Data Hiding**: Private fields and methods, getter and setter methods.

10. Advanced OOP Concepts

- **Static Members**: Static variables, static methods, and static blocks.
- **Final Keyword**: Final classes, methods, and variables.
- **This Keyword**: Referring to the current instance.
- **Super Keyword**: Accessing parent class members.
- **Nested and Inner Classes**: Member inner class, static nested class, local inner class, and anonymous inner class.

11. Packages and Import

- **Packages**: Creating and using packages, importing classes, `java.lang` package.

12. Exception Handling

- **Types of Exceptions**: Checked, unchecked, and errors.
- **Try-Catch Block**: Writing exception handling blocks.
- **Finally Block**: Ensuring resource release.
- **Throw and Throws**: Declaring exceptions and throwing exceptions.
- **Custom Exceptions**: Creating your own exception classes.

13. Collections Framework

- **Introduction to Collections**: Benefits and hierarchy of collections.
- **List Interface**: ArrayList, LinkedList, Vector, Stack.
- **Set Interface**: HashSet, LinkedHashSet, TreeSet.
- **Map Interface**: HashMap, LinkedHashMap, TreeMap, Hashtable.
- **Queue Interface**: PriorityQueue, Deque.
- **Collections Class**: Utility methods for collections.
- **Iterators**: Using Iterator and ListIterator.

14. Java Generics

- **Introduction to Generics**: Benefits and basic syntax.
- **Generic Classes**: Defining classes with generic types.
- **Generic Methods**: Creating methods with generic types.
- **Bounded Type Parameters**: Using `<T extends Class>` in generics.
- **Wildcard in Generics**: `? extends` and `? super` keywords.

15. Java I/O Streams

- **Introduction to Streams**: Byte streams vs. character streams.
- **File Handling**: File, FileReader, FileWriter, BufferedReader, BufferedWriter.
- **Serialization and Deserialization**: Using `Serializable` interface.
- **Object Streams**: ObjectInputStream and ObjectOutputStream.
- **NIO (New I/O)**: Overview of java.nio package, buffers, channels.

16. Multithreading and Concurrency

- **Introduction to Threads**: Creating threads using `Thread` class and `Runnable` interface.
- **Thread Lifecycle**: States of a thread, thread methods.
- **Synchronization**: Synchronized methods and blocks, static synchronization.
- **Inter-Thread Communication**: `wait()`, `notify()`, `notifyAll()`.
- **Thread Pools**: Executor framework.
- **Concurrency Utilities**: Locks, semaphores, countdown latches, concurrent collections.

17. Java 8 Features

- **Lambda Expressions**: Syntax and usage.
- **Functional Interfaces**: Predicate, Function, Consumer, Supplier.
- **Stream API**: Creating streams, intermediate and terminal operations, parallel streams.
- **Optional Class**: Handling null values.
- **Default and Static Methods**: In interfaces.
- **Method References**: Reference to a static method, instance method, and constructor.

18. Best Practices and Design Patterns

- **Java Coding Best Practices**: Naming conventions, code organization, and common pitfalls.
- **Design Patterns**: Singleton, Factory, Strategy, Observer, etc.

- **Code Optimization**: Profiling, memory management, garbage collection tuning.

19. Build Tools and Project Management

- **Maven**: Dependency management, building projects.
- **Gradle**: Build automation.
- **JAR Files**: Creating and using JAR files.

20. Testing in Java

- **JUnit**: Writing unit tests, annotations, assertions.
- **Mockito**: Mocking objects, writing test doubles.
- **Test-Driven Development (TDD)**: Writing tests before code.

21. Working with Databases

- **JDBC**: Connecting to databases, executing queries, handling results.
- **ORM Frameworks**: Introduction to Hibernate, JPA (optional for core).

22. IDE Features and Debugging

- **Using IDEs Effectively**: Code generation, debugging, refactoring.
- **Debugging Techniques**: Breakpoints, watches, step execution.

23. Version Control

- **Git Basics**: Cloning, committing, branching, merging.
- **Working with Git Repositories**: Pushing to and pulling from remote repositories.

24. Projects and Practice

- **Mini Projects:** Implement a small project, such as a library management system, bank application, or to-do list.
- **Contribute to Open Source:** Contributing to Java projects on GitHub.

Learning Resources

- **Books:** “Effective Java” by Joshua Bloch, “Java: The Complete Reference” by Herbert Schildt.
- **Online Courses:** Platforms like Udemy, Coursera, or YouTube.
- **Documentation:** Official Java Documentation (https://docs.oracle.com/en/java/).

This roadmap provides a comprehensive guide for mastering Core Java, from basic concepts to advanced features. As you progress, make sure to build projects, practice coding, and explore additional resources to deepen your understanding.

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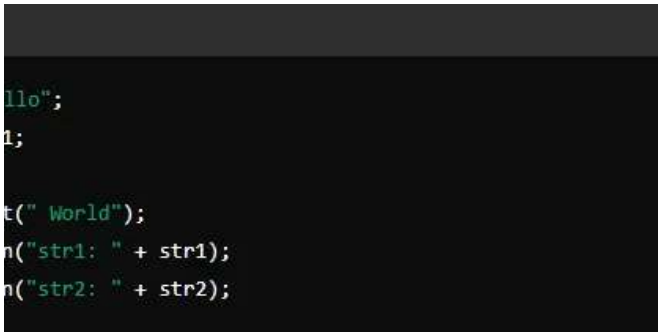

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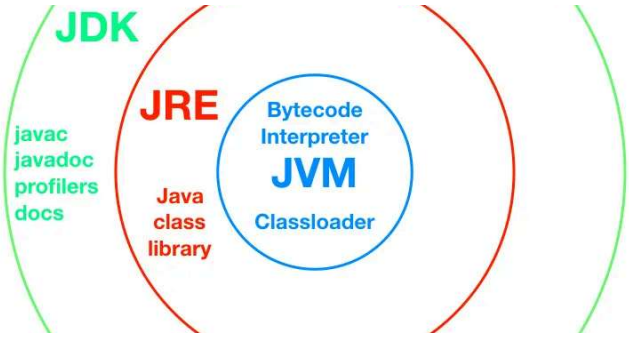

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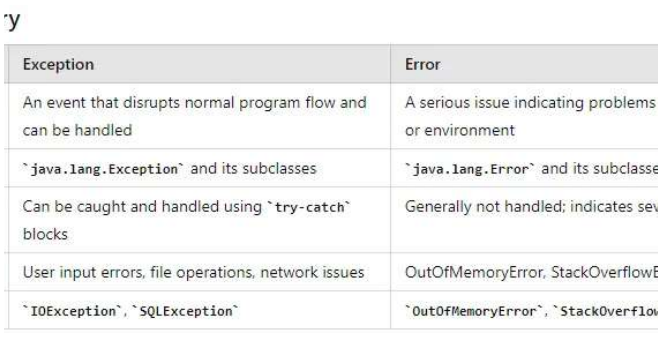

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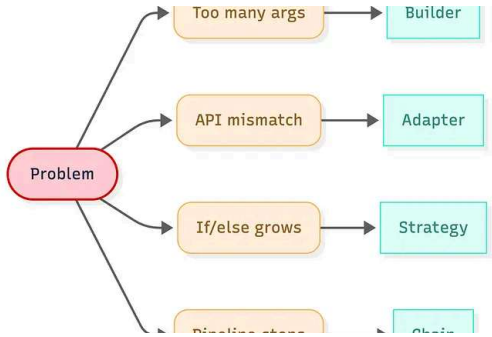

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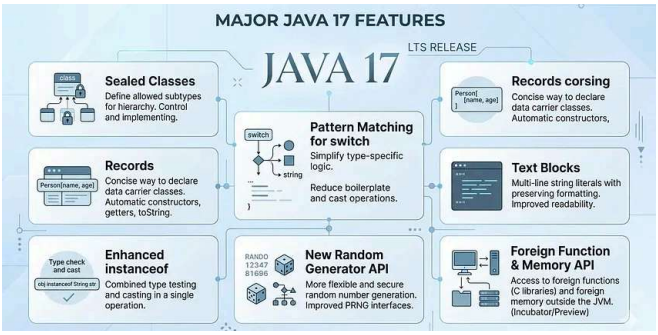

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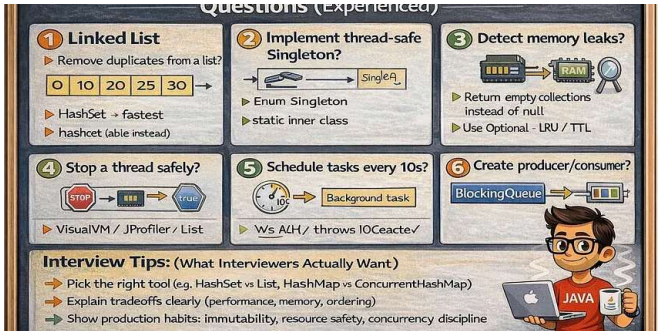

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